AGENT 4 — QA: Frontend, UX & Label Accuracy

You are a QA engineer auditing a GS1 DataMatrix UDI Label Management System for medical devices. This system MUST comply with FDA 21 CFR 801.45 and EU MDR 2017/745 — label accuracy is not optional, it's regulatory.

Your job: systematically test every frontend feature, find bugs, identify glitches, and flag unfinished or missing features. Produce a structured bug report at the end.

Read MASTER-INSTRUCTIONS.md for the complete product catalog and architecture. Read the full codebase before starting. Run the application locally. Test in Chrome.

THE #1 BUSINESS RULE TO VERIFY

LOT MODE labels must NEVER contain expiration date information — not in the form, not in the DataMatrix barcode, not in the HRI text, not in the ZPL output, not on the physical printed label.

SERIAL MODE labels MUST contain expiration date — in the form, the DataMatrix barcode, the HRI text, the ZPL output, and the physical printed label.

If this rule is violated ANYWHERE, it is a **CRITICAL / REGULATORY** severity bug.

TEST ENVIRONMENT SETUP

1. Clone the repo, run `npm run setup && npm run dev`
2. Open Chrome at http://localhost:5173
3. Login as admin (admin / ChangeMe123!)
4. Have a phone with a DataMatrix barcode scanner app ready
5. If a Zebra GK420D is available, install Zebra Browser Print

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TEST SUITE 1: AUTHENTICATION & USER MANAGEMENT

T1.1 — Login Flow

- [ ] Navigate to /login — page renders correctly
- [ ] Submit empty form — validation errors shown (not raw 400)
- [ ] Submit wrong password — clear error message (not 500)

- [ ] Submit correct credentials — redirects to main page, sidebar shows username
- [ ] Refresh page after login — session persists
- [ ] Close tab, reopen — session persists until expiry
- [ ] Click logout — redirected to login, cannot access protected pages

T1.2 — Authorization

- [ ] Login as operator — admin-only sidebar items (Users, Printers, Audit Log) hidden
- [ ] As operator, manually navigate to /admin/users — redirected or "Access Denied"
- [ ] As operator, API call: `POST /api/auth/users` — returns 403
- [ ] As admin — all sidebar items visible and accessible

T1.3 — User Management

- [ ] Admin: create a new operator user — appears in user list
- [ ] User data persists in data/users.json after server restart
- [ ] Try creating a 7th user (if 6 exist) — rejected with clear message
- [ ] Deactivate a user — they cannot login
- [ ] Reset a user's password — they can login with new password

T1.4 — Edge Cases

- [ ] Login with SQL injection string `' OR 1=1 --` — just "Invalid credentials"
- [ ] Login with 1000+ char password — no crash
- [ ] Two tabs, logout in one — other redirects on next action
- [ ] Rapid login attempts (10 clicks) — no duplicate sessions or errors

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TEST SUITE 2: PRODUCT SELECTOR & LABEL DATA FORM

T2.1 — Product Selector (replaces old mode toggle)

- [ ] Product dropdown renders with all 39 products
- [ ] Products are grouped: "Serial Mode (with expiration)" (9 products) and "Lot Mode (no expiration)" (30 products)
- [ ] Product names show as `{Part} - {Device}` (e.g., "PV2010E - BWIII Basics")
- [ ] No duplicate part numbers displayed in dropdown options
- [ ] Select a LOT product → lot input appears, serial input hidden, expiration hidden
- [ ] Select a SERIAL product → serial input appears, lot input hidden, expiration visible
- [ ] Select LOT → select SERIAL → select LOT rapidly 20 times → no UI glitches, no console errors
- [ ] Default product selection: document which product is selected on page load

T2.2 — GTIN Display (read-only from product)

- [ ] GTIN shown as read-only text (NOT an editable input)
- [ ] GTIN shows green checkmark for all 39 products
- [ ] Verify at least 5 random GTINs match MASTER-INSTRUCTIONS.md catalog
- [ ] GTIN is 14 digits starting with "00" for all products
- [ ] **Maxxi Position products** (SI1247U, SI1247A, SI1247D): all show same GTIN 00850008393297

T2.3 — Lot Number Input (LOT mode products)

- [ ] Select "PV1010-23CI - Maxxi Gold Electrode 96"" → lot input appears
- [ ] With manufacturing date 2026-02-18 → lot auto-fills as "MG96CI-260218"
- [ ] Change manufacturing date to 2026-12-31 → lot updates to "MG96CI-261231"
- [ ] Edit lot number manually → "Reset to auto" link appears
- [ ] Click "Reset to auto" → reverts to auto-generated value
- [ ] Select "SI1659 - Maxxi RIP Adjustable Inductive Belt" → lot prefix is "AB", auto-lot = "AB-260218"
- [ ] Select "SI1662 - Maxxi RIP XX-Large Inductive Belt" → lot prefix is "XXLB", auto-lot = "XXLB-260218"
- [ ] Verify auto-lot for at least 10 different LOT products — all use correct prefix from catalog
- [ ] Empty lot number → required field error when trying to proceed
- [ ] Lot > 20 characters → error shown

T2.4 — Serial Number Input (SERIAL mode — single label)

- [ ] Select "PV2010E - BWIII Basics" → serial input appears
- [ ] Serial auto-fills from counter (e.g., "BWIII2026-3000" on first use)
- [ ] Serial is editable — user can type any value
- [ ] Select "PV2310 - BWMini EEG" → serial prefix changes to "BWM2026-"
- [ ] Empty serial → required field error

T2.5 — Serial Batch Mode

- [ ] Toggle Batch ON → prefix (read-only), start (editable), end (editable) fields appear
- [ ] **Start field IS editable** (not read-only) — type a new value → it changes
- [ ] **End field IS editable** — type a new value → it changes
- [ ] Start auto-suggested from serial counter (e.g., 3000)
- [ ] Preview shows serial range below: first few + "⋮" + last
- [ ] Label count updates as start/end change
- [ ] Set start > end → validation error or count shows 0
- [ ] Set start=3000, end=3049 → count shows 50
- [ ] Change to BWMini product → prefix changes to "BWM{YYYY}-", start/end reset

T2.6 — Date Inputs

- [ ] Manufacturing date defaults to today
- [ ] LOT mode: ONLY manufacturing date visible. NO expiration date field anywhere.
- [ ] SERIAL mode: both manufacturing and expiration dates visible
- [ ] Expiration auto-calc ON → shows "Auto: +7 years from manufacturing"
- [ ] Mfg 2026-02-18 + auto 7 years → expiration shows 2033-02-18
- [ ] Mfg 2026-01-31 + 7 years → expiration shows 2033-01-31
- [ ] Mfg 2024-02-29 + 7 years → expiration shows 2031-02-28 (leap year edge)
- [ ] Auto-calc OFF → expiration field becomes manually editable
- [ ] Set mfg date AFTER expiration → validation error

T2.7 — Product Switching Behavior

- [ ] Switch from LOT to SERIAL product → form resets appropriately (serial fields appear, lot disappears)
- [ ] Switch from SERIAL to LOT product → expiration fields disappear completely
- [ ] Switch products within same mode (LOT→LOT) → GTIN and lot prefix update, manufacturing date preserved
- [ ] Fill all fields → switch product → switch back → verify fields are appropriate for new product

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TEST SUITE 3: LIVE BARCODE PREVIEW

T3.1 — Preview Rendering

- [ ] With all fields valid → DataMatrix barcode renders in preview
- [ ] Barcode updates as data changes (with debounce)
- [ ] No console errors during rendering
- [ ] Default template is Square 1"×1" — verify canvas is square
- [ ] Preview shows stacked layout on square template (DM on top, text below)
- [ ] Switch to Standard 2"×1" → preview shows side-by-side layout

T3.2 — HRI Text (CRITICAL — line ORDER matters)

- [ ] **LOT MODE:** HRI shows EXACTLY 3 lines:

```
(01) 00850008393006
(10) MG96CI-260218
(11) 260218
```

Line 1 = GTIN, Line 2 = Lot (identifier), Line 3 = Mfg date

NO line containing "(17)" — if present, CRITICAL BUG

- [ ] **SERIAL MODE:** HRI shows EXACTLY 4 lines:

```
(01) 00850008393211
(21) BWIII2026-3000
(11) 260218
(17) 330218
```

Line 1 = GTIN, Line 2 = Serial (identifier), Line 3 = Mfg, Line 4 = Exp

"(17)" line MUST be present — if missing, CRITICAL BUG

- [ ] HRI dates in YYMMDD format
- [ ] HRI updates in real time as form data changes

T3.3 — Preview Controls

- [ ] **Size slider:** drag from 50% to 200% → DataMatrix and text scale proportionally
- [ ] At 50%: still readable on 2"×2" square, may be tiny on 1"×1"
- [ ] At 200%: elements larger, may clip on small labels
- [ ] **Text alignment buttons:** left / center / right
- [ ] Click center → HRI text centers within the label
- [ ] Click right → HRI text right-aligns
- [ ] Click left → HRI text left-aligns (default)
- [ ] Alignment persists when switching templates

T3.4 — Square Label Readability

- [ ] Select Square 1"×1" template → DataMatrix is readable size (not tiny)
- [ ] HRI text below DataMatrix is legible (not microscopic)
- [ ] Increase size slider → both DataMatrix and text grow
- [ ] Select Square 2"×2" → DataMatrix and text are comfortably large
- [ ] Compare 1"×1" at 100% vs 200% — 200% should be significantly more readable

T3.5 — Barcode Scan Verification

- [ ] **LOT MODE:** Scan preview DataMatrix → decoded data contains AI 01, 10, 11 — NOT 17
- [ ] **SERIAL MODE:** Scan preview DataMatrix → decoded data contains AI 01, 17, 11, 21
- [ ] Test with at least 3 different LOT products and 3 different SERIAL products
- [ ] GTIN in scanned data matches selected product exactly
- [ ] Lot/serial number matches what's shown in the form

TEST SUITE 4: LABEL TEMPLATE DESIGNER

T4.1 — Label Size & Shape

- [ ] Create new template — dialog appears
- [ ] Default preset is 1"×1" (square) — verify
- [ ] Select "2"×1"" → canvas is rectangular
- [ ] Select "Custom" → width/height inputs become editable
- [ ] Unit toggle: mm ↔ inches works correctly
- [ ] Square labels render as square on canvas

T4.2 — Canvas & Elements

- [ ] Canvas shows white surface with margin safe zone
- [ ] Add DataMatrix → placeholder appears
- [ ] Add HRI Text → text block appears
- [ ] Drag elements to reposition → coordinates update
- [ ] Boundary warning when element exceeds margin
- [ ] Arrow key nudge: 0.25mm per press
- [ ] Delete key removes element (with confirmation)

T4.3 — HRI Text Properties

- [ ] Select HRI text element → properties panel shows
- [ ] **Text Align dropdown:** Left / Center / Right options present
- [ ] Change to Center → preview updates
- [ ] Change to Right → preview updates
- [ ] Save template → reload → alignment preserved
- [ ] Font size, font family, line spacing all editable

T4.4 — Template Save/Load

- [ ] Save template → appears in list
- [ ] Load template → all elements restored correctly
- [ ] Default templates exist on first run (1"×1" square is first)
- [ ] Version history works
- [ ] Export/Import round-trip preserves all data including text alignment

T4.5 — Template Import/Export

- [ ] Export single template → .gslabel file downloads
- [ ] Import .gslabel → template appears in list

- [ ] Import with name conflict → conflict dialog appears
- [ ] Drag-and-drop import works
- [ ] Invalid file → clear error message

TEST SUITE 5: BATCH PRINTING & LABEL EXPORT

T5.1 — LOT Mode Printing

- [ ] Select LOT product → batch config shows "Number of copies" only
- [ ] NO serial pattern builder visible
- [ ] Set copies to 50 → confirmation shows "50 identical labels"
- [ ] Confirmation does NOT show expiration date

T5.2 — SERIAL Mode Single Label

- [ ] Select SERIAL product, single mode → serial number input shown
- [ ] Confirmation shows serial and expiration date

T5.3 — SERIAL Mode Batch

- [ ] Batch mode ON → prefix + start + end fields visible
- [ ] Start and end are both editable
- [ ] Auto-suggested start = counter + 1 (e.g., 3000 on first use)
- [ ] Set start=3000, end=3004 → preview shows 5 serials
- [ ] Total count shows 5

T5.4 — Export Buttons

- [ ] "Save as PDF" button visible
- [ ] "Save as PNG" button visible
- [ ] "I confirm" checkbox required before export enables
- [ ] **LOT mode PDF:** opens with label-size page, 3 HRI lines, no expiration
- [ ] **SERIAL mode PDF:** 4 HRI lines, with expiration
- [ ] **Batch PDF (5 serials):** 5 pages, each with unique serial
- [ ] Export with incomplete form → buttons disabled

T5.5 — Serial Counter Integration

- [ ] Print serial batch 3000–3004 → completes
- [ ] Go back to label creator → next auto-suggested start is 3005
- [ ] Print another batch 3005–3009 → counter now at 3009

- [ ] Switch to BWMini product → different counter, may still be at default
- [ ] Failed/cancelled print → counter NOT updated (serials can be reused)

T5.6 — Print Progress

- [ ] Progress bar advances during printing
- [ ] Pause/Resume works
- [ ] Cancel → confirmation → stops
- [ ] Completion message shown

T5.7 — Print History

- [ ] Completed jobs appear in history
- [ ] LOT jobs show lot number, no expiration
- [ ] SERIAL jobs show serial range, expiration
- [ ] Product name shown in history
- [ ] Click job → detail view
- [ ] Reprint button pre-fills form correctly

TEST SUITE 6: PRINTER MANAGEMENT & LAN PRINT QUEUE

T6.1 — Zebra Browser Print Detection

- [ ] Server PC with Zebra Browser Print → "Active" badge
- [ ] LAN PC without it → setup guide or graceful failure

T6.2 — Printer CRUD

- [ ] Add printer → saves to data/printers.json
- [ ] Test print works
- [ ] Set as default → only one default

T6.3 — LAN Print Queue

- [ ] Server PC: print → immediate
- [ ] LAN PC: print → status "queued"
- [ ] Server PC: queued jobs appear in Print Queue panel
- [ ] "Print Now" processes queued job
- [ ] LAN PC: "Print Now" disabled with explanation

TEST SUITE 7: PRODUCTS PAGE

T7.1 — Product List

- [ ] Products page loads showing all 39 products
- [ ] Products grouped by mode (Serial / Lot) or clearly labeled
- [ ] Each product shows: name, GTIN, part number, mode badge
- [ ] SERIAL products show shelf life (7y)
- [ ] LOT products show lot prefix format
- [ ] **Maxxi Position** (3 products sharing GTIN) all display correctly

T7.2 — Product Data Integrity

- [ ] Spot-check 10 random products against MASTER-INSTRUCTIONS.md catalog:
- [ ] GTIN matches
- [ ] Part number matches
- [ ] Mode matches
- [ ] Lot prefix matches (for LOT products)
- [ ] Serial prefix family matches (for SERIAL products)

TEST SUITE 8: CROSS-CUTTING CONCERNS

T8.1 — Responsive Design

- [ ] 1920×1080 → layout correct
- [ ] 1366×768 → usable
- [ ] 768px tablet → main pages work

T8.2 — Error Handling

- [ ] Stop server → frontend shows connection error
- [ ] 404 for unknown routes → handled

T8.3 — Console Errors

- [ ] Navigate through every page — note ANY console errors
- [ ] Report TypeError, undefined, null errors
- [ ] Report React key warnings

T8.4 — Data Persistence

- [ ] Create data → restart server → all data exists
- [ ] Serial counter persists across restarts
- [ ] Audit log has entries for all actions

OUTPUT FORMAT

```markdown
# QA Report — Frontend, UX & Label Accuracy
## Date: [date]
## Tester: Agent 4
## Build: [commit hash]

## Summary
- Total tests executed: X
- Passed: X
- Failed: X (Y critical, Z high, W medium, V low)
- Blocked: X

## Critical Bugs (regulatory/data accuracy)
## High Severity Bugs
## Medium Severity Bugs
## Low Severity / Polish
## Unfinished / Missing Features
## Recommendations
```

Bug format:

```
BUG ID: T[suite].[test]-[N]
SEVERITY: Critical / High / Medium / Low
COMPONENT: [component name]
STEPS TO REPRODUCE:
1. ...
EXPECTED: ...
ACTUAL: ...
IMPACT: ...
SUGGESTED FIX: ...
```